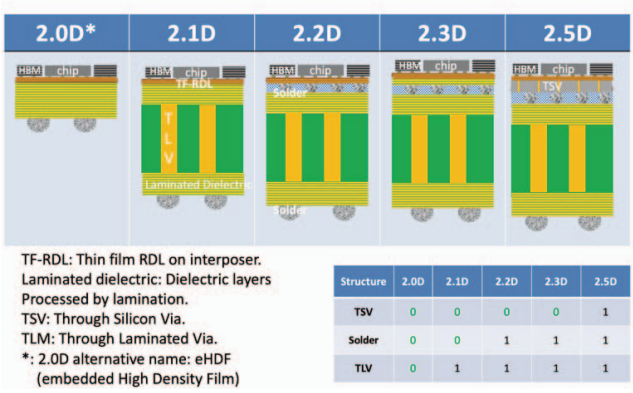
solders in the package would be beneficial to the electrical performance of the system. Solder-less solutions like FO-WLP, InFO-WLP and embedded die have been in production for many years.

There are various trade names used in the HPC packaging structure; namely 2.5D, SLIT, and CoWoS-S. It is desirable to classify the HPC packaging by their structure. Fig. 2 classifies HPC packaging by the components inside their structure, with increasing structure complexity from 2.0D constitute to to 2.5D. For the 2.5D structure, silicon interposer with TSVs are connected to the underlayer cored substrate with solders. The cored substrate has TLVs (Through Laminated Vias). On the other hand, the 2.0D packaging is a TSV-less, TLV-less and solder-less structure that can take the full benefit of the system electrical performance. The 2.0D TV's manufacturability and reliability have been verified by SiPlus and Nanya PCB Co. [7].

The advanced 2.0D structure that combines BEOL, thin film RDL (TF-RDL) and laminated dielectric layers (ABF, PP) is constituted to one "integrated substrate". This integrated substrate can be used for the die last chiplets and dielets applications in Fig. 3 [8].

RDL interposer (Amkor/Samsung) and CoWoS-R (tsmc) are similar to the 2.2D structure after chip assembly. However, those are "die middle" and not a true die last solution. Since dies are bonded to RDL and then mold, de-bond before joining to the substrate. It is desirable to have a substrate that is a true 2.2D die last solution.

In this paper, an innovation 2.2D die last solution to enable TF-RDL directly on substrate is proposed. Test vehicles have been built to test its feasibility and manufacturability.



Structure	2.0D	2.1D	2.2D	2.3D	2.5D
TSV	0	0	0	0	1
Solder	0	0	1	1	1
TLV	0	1	1	1	1

Fig. 2. Classification of HPC according to their packaging structure. Note that from 2.0D, 2.1D to 2.5D with increasing packaging structure complexity.

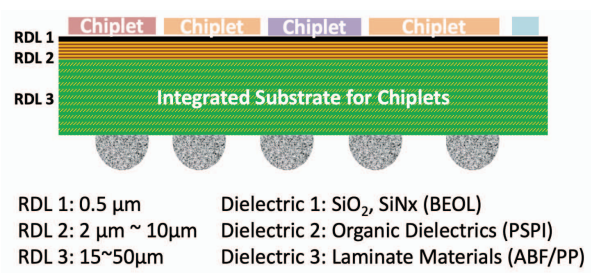


Fig. 3. Advanced 2.0D structure that combines BEOL, TF-RDL and laminated dielectric into one high performance 2.0D substrate that can be used for dielets and chiplets applications.

II. 2.2D Test Vehicle Design

To test the feasibility of the 2.2D structure, three important items need to be verified. (1) The dimensional stability of the TF- RDL needs to be maintained, otherwise there will be mis-registration between thin film RDL and substrate. (2) The robustness of the joints between TF-RDL and substrate need pass the substrate reliability test. (3) It is better to take the advantages of the existing infrastructure to realize the low cost manufacture.

There are several joint structures and surface metallization between TF-RDL and substrate are possible. For example, pillar connects to pad with solder on top, pillar to pillar connection with solder, or Cu to Cu direct bonding etc. There are also several surface metallization options, such as Cu to Cu, Au to Au, Au to solder, Au to Sn, etc. Furthermore, there are options for substrate material selections, for example, organic, ceramics and glass substrate are under consideration.

In this study, copper pad with solder plating is the structure at the TF-RDL side and copper pillar with Au surface finish on ceramic is at the substrate side.

To check the connectivity of TF-RDL directly to the substrate, 3x3mm serpentine loops on TF-RDL and on ceramic substrate are designed. Fig. 4 shows the CAD design pattern of the TF-RDL. The thin film pads with diameter of 400µm and pitch 500µm are designed. 5µm Asahi Kasei low temperature curing Polyimide are used as thin film dielectric layer. 3µm Copper metallization are used for conduction layer with Ti/Cu seed layer underneath. There is a lead free solder plating on top of each TF-RDL pads.

AlN ceramic substrate with 1mm thickness was chosen as a substrate. To construct the vertical serpentine line, pillars on the AlN substrate are designed to be connected in pairs (Fig. 5).

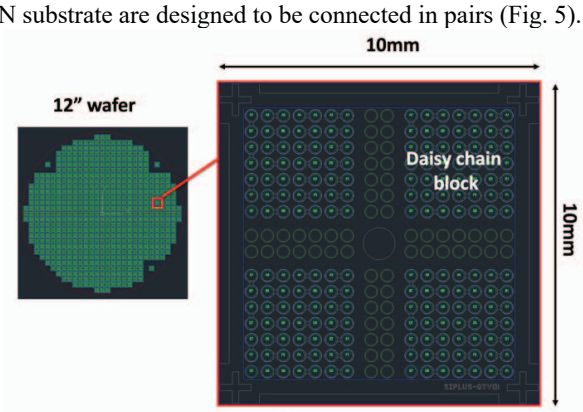


Fig. 4. The thin film CAD test pattern designs.

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